

For each of the following draw a force diagram:

1. Find the acceleration produced when a body of mass 5 kg experiences a resultant force of 10 N.
 2. Find the resultant force that would give a body of mass 3 kg an acceleration of 2 ms^{-2} .
 3. A humpback whale of mass 145 kg accelerates at $\frac{1}{2} \text{ ms}^{-2}$. What resultant force is required to produce this acceleration?
 4. The driving force on a Peugeot 206 is 1500 N. There is a frictional resistance to the motion of 800 N. The Peugeot has a mass of 1400 kg.
 - (a) Draw a diagram labelling the horizontal forces on the Peugeot and the acceleration.
 - (b) What is the resultant force on the Peugeot?
 - (c) What is the acceleration of the Peugeot?
 5. A Smart car of mass 730 kg moves along a level road with an acceleration of 2 ms^{-2} . If its engine is exerting a forward force of 1700 N, what resistance is the car experiencing?
 6. A bicycle of mass 90 kg (including the cyclist) is travelling at a constant velocity of 6 ms^{-1} . If the forward driving force from the ground on the back wheel is 200 N, what is the frictional resistance to the motion of the bicycle?
 7. The frictional resistance from the air and ground on a milk float of mass 1200 kg is 50 N. The milk float applies its brakes which give a braking force of 200 N. Draw a diagram labelling the horizontal forces and acceleration. What is the deceleration of the milk float?
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8. A box of mass 5 kg is lowered vertically by a rope. Draw a labelled diagram and find the tension in the rope when the box is lowered with an acceleration of 4 ms^{-2} .
 9. The acceleration due to gravity on the moon is approximately 1.6 ms^{-2} . For a Sixth former of mass 60 kg, find his:
 - (a) weight on Earth
 - (b) mass on the Moon
 - (c) weight on the Moon
 10. A pile of bricks of total mass 100 kg is hoisted up the side of a house. Draw a labelled diagram and find the force in the lifting rope when the bricks are lifted with an acceleration of $\frac{1}{4} \text{ ms}^{-2}$.
 11. A grand piano of mass 700 kg is at rest and attached to a chain hoist. If the chain pulls upwards on the piano with force 7000 N, find the piano's acceleration.
 12. A stone of mass 50 g is dropped into some liquid and falls vertically through it with an acceleration of 5.8 ms^{-2} . Find the force of resistance acting on the stone.

13. A body of mass 10 kg is attached to a vertical string. Draw a diagram labelling the weight of the body and the tension in the string, T . Find the force T in each of these situations:
 - (a) the string raises the body with an acceleration of 5ms^{-2}
 - (b) the string lowers the body with an acceleration of 5ms^{-2}
 - (c) the string raises the body at a constant velocity of 5ms^{-1}
 - (d) the string lowers the body at a constant velocity of 5ms^{-1}
14. I push the book on my desk with my finger, exerting a horizontal force of 25 N on the book. The book does not move. (a) What is the frictional force on the book from the table? (b) I now push with a force of only 15 N on the book. What happens to the book? What is the frictional force on the book now?
15. At one point during its plummet to Earth, a meteorite experiences an upward force due to air resistance of 5000 N. If its acceleration downwards at this moment is 2ms^{-2} , what is the meteorite's mass?
16. A miners' cage of mass 420 kg carries three miners of average mass 80 kg. On leaving the bottom of the mine the cage accelerates upwards at 2ms^{-2} for the first few seconds of motion. Find the force in the cable during the first few seconds of motion.
17. I let go of a book of mass 700 g. Taking air resistance to be negligible, what is the book's acceleration?
18. A flea of mass 0.08 g accelerates at 1500ms^{-2} during take-off. What resultant force is required to produce this acceleration?
19. A bucket has a mass of 5 kg when empty and 15 kg when full of water. The empty bucket is lowered into a well, at a constant acceleration of 5ms^{-2} , by means of a rope. When full of water the bucket is raised at a constant velocity of 2ms^{-1} . Neglecting the weight of the rope, find the force in the rope (a) when lowering the bucket; (b) when raising the full bucket.
20. A lift of mass 500 kg is drawn up by a cable. It makes an ascent in three stages: it is brought from rest to its maximum speed by a constant acceleration of $\frac{g}{2}\text{ms}^{-2}$, it then moves with its maximum speed for an interval of time and is then brought to rest by a deceleration of $g\text{ms}^{-2}$. Find the tension in the cable in terms of g in each of the three stages.